

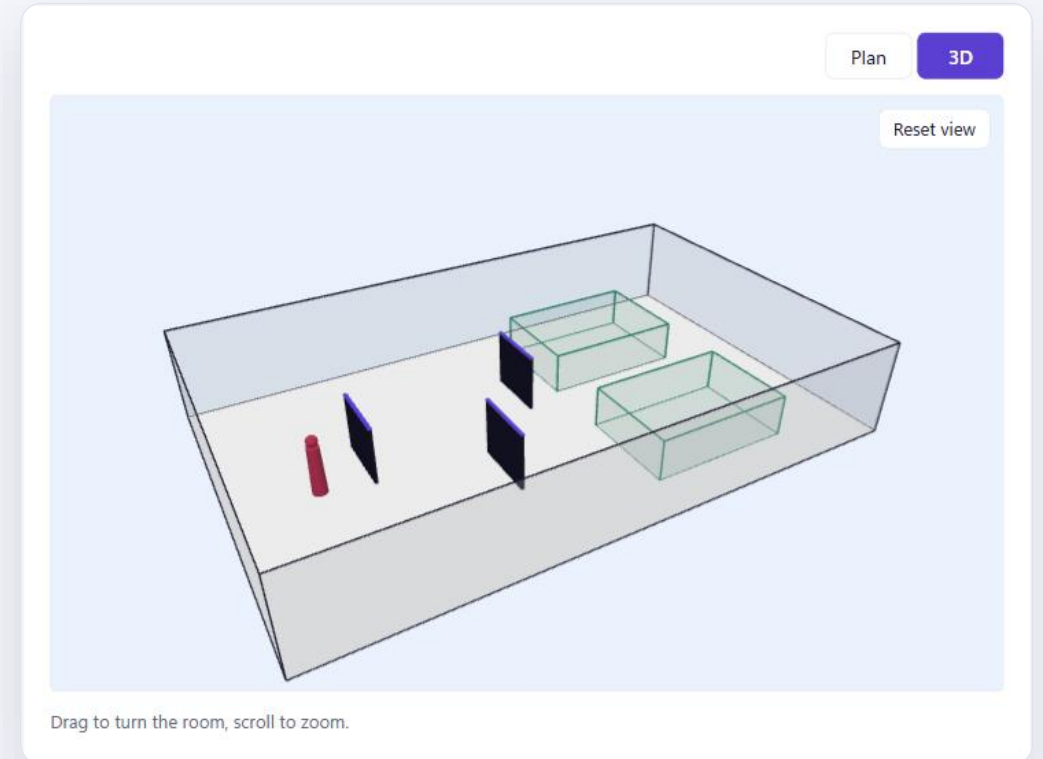


Powered by Quanscient Allsolve

Allquiet

A quieter workplace is a healthier one. Physics shows where the panels go.

Veli Lampila · Santeri Hukari · Arman Borujerdi



NOISE IS A HEALTH ISSUE

Among environmental causes of ill health in Europe, noise is **second only to air pollution**.

Traffic noise alone costs western Europe more than 1 million healthy life-years every year. Then we go to work and sit in a shared room for eight hours.

THE EVIDENCE

What office noise does to people

+34%

sweat response, a physical sign of stress, after 8 minutes of open-plan noise

Sander et al., 2021

+25%

negative mood in the same experiment

Sander et al., 2021

−40%

attempts at a hard puzzle after 3 hours of low-level office noise, with raised adrenaline

Evans & Johnson, 2000

In a survey of more than 42,000 office workers, noise and lost sound privacy outweighed the benefit of easier interaction in open-plan offices.

THE GAP

Offices buy acoustic panels, then place them **by eye**.

If you can afford only three panels, where do they protect the most people?

A rule of thumb gives a direction. Nobody gets a position, or a number for what the second and third panel are worth.

WHY IT IS HARD

Speech is as big as the **furniture**

Speech in air

Frequency	Wavelength
250 Hz	1.37 m
500 Hz	0.69 m
1 kHz	0.34 m
2 kHz	0.17 m

Things in an office

Object	Size
Acoustic screen	1.2–2.6 m
Desk	1.6 m
Gap, screen to wall	1–2 m

Sound does not stop at a screen. It bends around the ends, and the bent waves interfere. Formulas and ray tracing are weakest exactly here, so the room has to be solved as a wave problem.

THE PRODUCT

From a room to a **placement**

1 Bring the room

Scan it with a phone and trace the walls, or draw the outline.

2 Mark noise and quiet

Where people talk, and where people need to concentrate.

3 Say how many panels

One to four, with a quick or a detailed search.

4 Get the positions

Allsolve simulates the layouts. Each panel comes back measured from the walls.

Plan3D

Add noise sourceAdd quiet zoneDelete

2. Noise and quiet zones

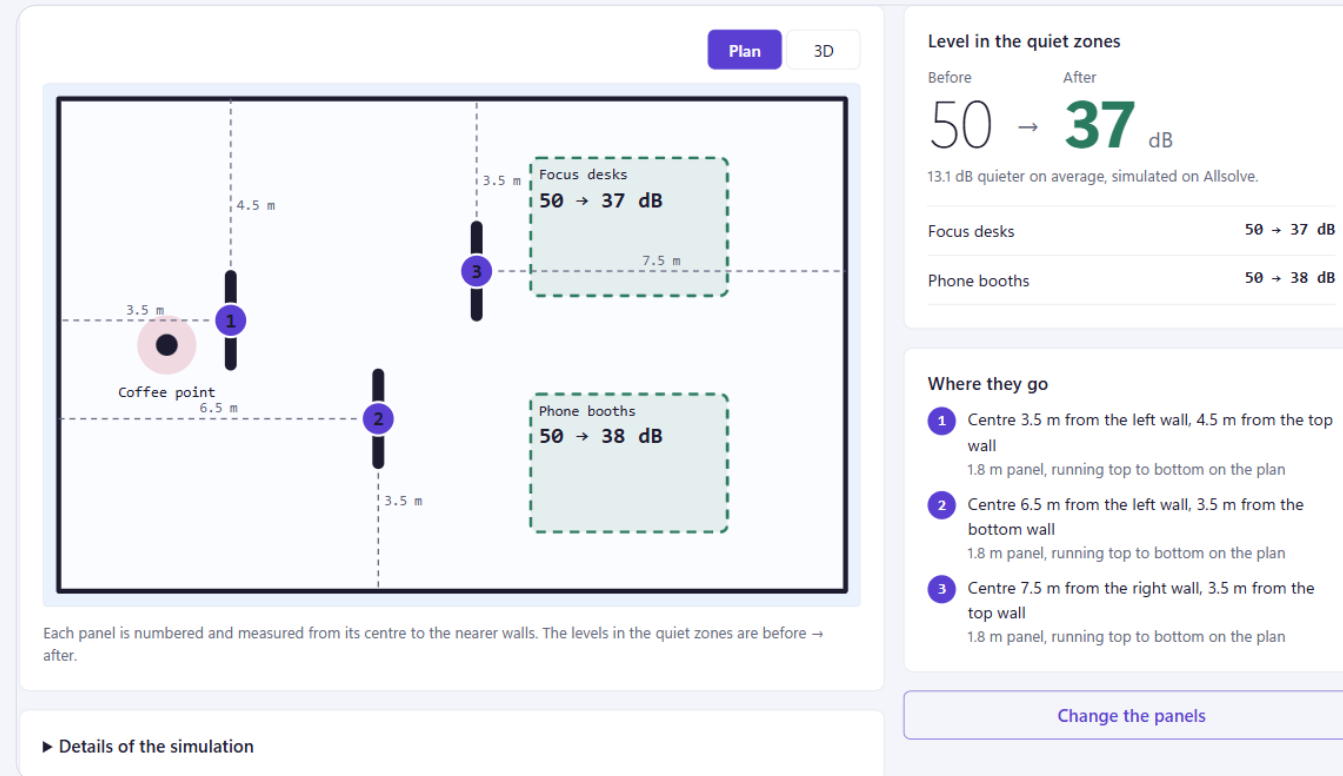
Noise sources	1
Quiet zones	2

Select a noise source to set how loud it is. Drag the corner of a quiet zone to resize it.

Next: panels

THE DEMO

Three panels, placed by **simulation**



50 → 37

dB in the quiet zones, before and after: 13 dB quieter on average

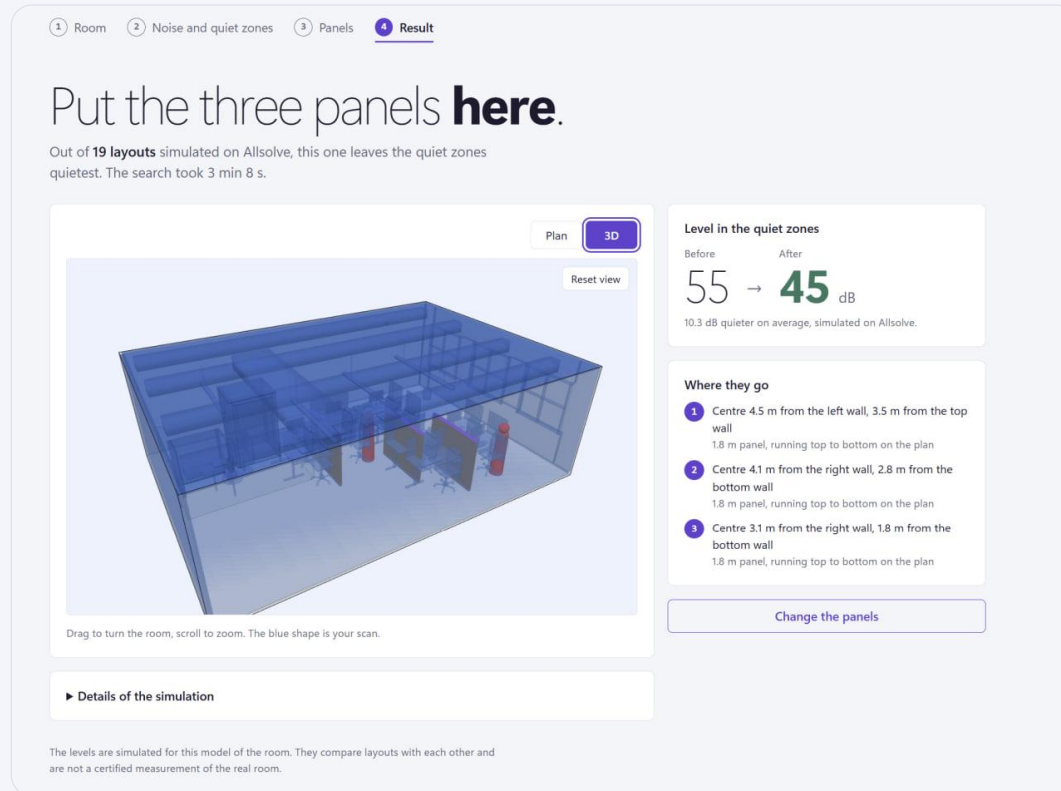
22 layouts and 44 wave solves on Allsolve in **33 seconds**

The in-browser rule of thumb scored this layout 78. Allsolve scored it 24, where 100 is the room with no panels.

Simulated levels for this model of the room. They compare layouts and are not a certified measurement.

A REAL ROOM

An office scanned with a phone



55 → 45

dB in the quiet zones, before and after three panels: 10 dB quieter on average

A 10.6 × 7.3 m office, traced over its own scan. **19 layouts** on the detailed search in **3 min 8 s**.

The blue shape is the scan. The panels come back as distances from the walls, ready for whoever carries them in.

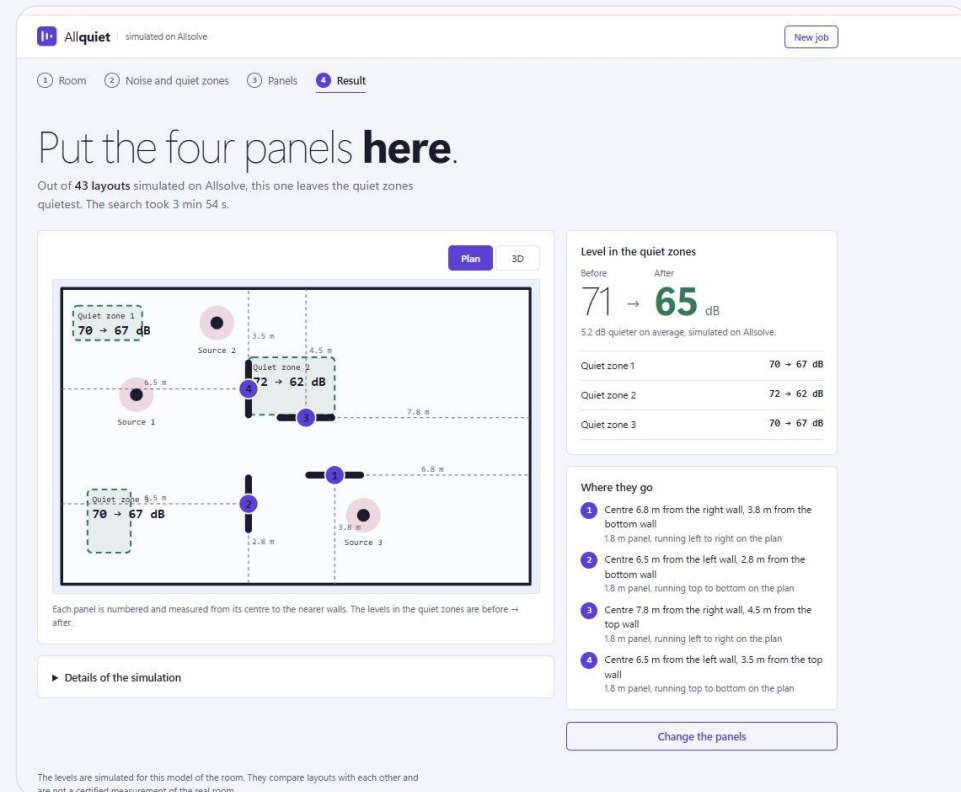
Simulated levels for this model of the room. They compare layouts and are not a certified measurement.

A HARDER ROOM

Three noise sources, three places to protect

72 → 62

dB in the worst quiet zone. The other two go from 70 to 67 dB.



Four panels, 43 layouts simulated in 3 min 54 s.

With noise from three sides, no guess puts four panels well. The search spends them where the people sit closest to the noise, and shows what each zone gains.

Simulated levels for this model of the room. They compare layouts and are not a certified measurement.

POWERED BY ALLSOLVE

A search needs hundreds of solves

100

cloud machines can run one sweep
at the same time

28 s

for 360 layouts: 720 wave solves
over two speech bands

24 s

for 803 layouts when the room's
meshes are used again

Panel positions are project variables in the Allsolve Python SDK. One project describes every layout, and the pressures at each listening point come back as numbers in Python.

Timings measured in this project on the 2D model of a 16 × 10 m office

WHILE IT RUNS

One click, 36 machines in the cloud

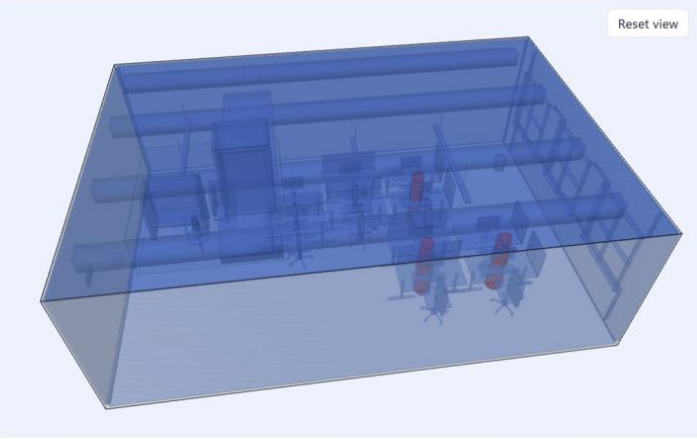
How many panels?

Choose the panels you have. The app simulates the room with them in different places and keeps the quietest layout.

Plan

3D

Reset view



Drag to turn the room, scroll to zoom. The blue shape is your scan.

► Details of the simulation

How many panels

1

2

3

4

Search

Quick
about half a minute

Detailed
several minutes

A top-down model of the room, 19 layouts. Each panel is placed in its own round, and every round waits for Allsolve to mesh and then to solve.

Stop

1:52 Meshing and solving 6 layouts (panel 2 of 3): 36 solves, all at the same time...

Allsolve project: [Allquiet 2D - 10.6 x 7.3 m - 3 panels - 03 Oct 14:18:29](#)

On Allsolve now: 36 machines running at the same time

Simulation (panel 2 of 3) running • 50%
36 machines

Status and counts are read from Allsolve each time this page asks for progress.

▼ Simulation settings

36

Allsolve machines solving one round of layouts at the same time

The app reads status and counts from Allsolve and links to the project, so every number can be traced.

THE ALTERNATIVES

Why not a simpler tool

Approach	Good at	Where it falls short for placement
Rules of thumb	Rating an existing office	They describe a result and cannot predict a layout
Barrier formulas	Instant, runs in a browser	One screen in open space: no interference, no room
Ray tracing	Large rooms, high frequencies	Weak at bending around screens one wavelength wide
Desktop FEM	The same wave physics	A licence and workstation per user, hard to script 200 runs
Allsolve + SDK	Wave physics in the cloud	Minutes, not milliseconds: we show an estimate first

Honest limits, clear **next steps**

What the number means today

- It ranks layouts in a simplified model. It is not a certified dB reduction.
- It covers the low and middle speech bands, where bending decides the outcome.
- The full 3D room works and costs more: two layouts took about five minutes.

What comes next

- Show the solver's own pressure field as the sound map.
- Absorbing panels and wall materials, not only blocking ones.
- Real floor plans through CAD import.
- A budget in euros in place of a panel count.



Quiet is a health measure.

Allquiet puts acoustic treatment where physics says it matters. Simulated on Quanscient Allsolve.

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